

Building a Derived Parcel-Level Target and Operational Datasets for Forecasting

Phase 4bis & 4ter — Methodological Report

Predicting the optimal harvest window of winter wheat
Centre-Val de Loire region, France (2020–2024)

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Educational document — plain-language version

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1. Why this work? The problem in plain language

A wheat farmer faces a simple but crucial question: **exactly when should the crop be harvested?** Harvest too early and the grain is too moist (costly drying is required). Harvest too late and the grain degrades, may sprout on the ear, or be damaged by rain. The “right” date varies from field to field depending on soil, exposure and local weather.

Our goal is to **predict this date in advance** for each of the 1,500 wheat parcels we study in the Centre-Val de Loire region, using only freely available data: satellite imagery, weather, and soil characteristics.

■ REAL EXAMPLE — An analogy

Imagine predicting when a runner will finish a marathon without watching them cross the finish line. You could observe their pace at the 30 km mark, the weather, their past training, and make an estimate. This is exactly our approach: we observe the “early warning signs” of wheat maturity (colour change, drying) captured by satellites, and infer the harvest date.

This report documents one specific step of the project: building a **target** (the value the model will learn to predict) and preparing several datasets for machine learning. It is a technical step, but we explain it step by step, without unnecessary jargon.

2. Two ways to define “the harvest date”

To train a model, we must first provide the “correct answer” to learn — the so-called **target**. We have two very different ways of defining the harvest date.

The REGIONAL target (official)

An official French source, **Céré’Obs** (run by FranceAgriMer), publishes weekly the percentage of wheat harvested in each region. By finding the week when this percentage crosses 50%, we obtain an “average” harvest date for the whole region. It is reliable and official, **but it gives only one date per year** for the entire Centre-Val de Loire region.

The DERIVED parcel-level target (our innovation)

The problem with the regional target: if all parcels in a given year share the same target date, the model learns little useful — it just learns “in 2021, harvest is on July 18”. It cannot distinguish an early parcel from a late one. This is why we built a **derived target**: a date estimated *specifically for each parcel*, from its own satellite signals.

Aspect	REGIONAL target	DERIVED target
Number of values	5 (one per year)	142 (per parcel)
Granularity	Whole region	Each parcel
Origin	Céré’Obs (official)	Satellites (computed)
Ground truth?	Yes	No (estimated proxy)
Internal variability	None	≈ 9 days

Table 1 — Comparison of the two target definitions.

■ KEY TAKEAWAY

We keep BOTH targets. The regional target is a solid, reliable reference (our “baseline”). The derived target is more ambitious and is the thesis innovation. Comparing them is itself a scientific contribution.

3. How our two “sensors” work (plain language)

The derived target relies on two satellite types that “see” the plant in two complementary ways. Understanding them is essential to understanding our choices.

3.1 Optical satellite (Sentinel-2): plant colour

Sentinel-2 photographs the Earth in several colours, including ones invisible to the human eye (infrared). From these colours we compute the **NDVI**, an index measuring “how green and active the vegetation is”. A healthy plant has a high NDVI (close to 0.9); a dry plant or bare soil has a low NDVI (close to 0.2).

■ REAL EXAMPLE — The wheat cycle seen by NDVI

In winter, young wheat has a moderate NDVI. In spring it rises (the plant grows and greens) to a peak around April-May. Then, as harvest approaches, wheat yellows and dries: NDVI **drops sharply**. This drop is called **senescence**. It is an excellent indicator: harvest usually comes a few days to two weeks after senescence begins.

The main drawback of optical data: **clouds block the view**. When clouds are present, the satellite cannot see the ground, and observations are lost. This is why we need a second sensor.

3.2 Radar satellite (Sentinel-1): plant structure

Sentinel-1 uses radar: it sends a wave toward the ground and measures the returning echo. Huge advantage: **radar sees through clouds** and works day and night. It does not see colour, but the **structure** and **moisture** of what is on the ground.

Radar measures two “polarisations” called **VV** and **VH**. Without going into the physics, remember that their ratio (the **VV/VH ratio**) describes vegetation structure. When wheat is dense and tall, this ratio is low. When wheat is cut and bare soil reappears, the ratio **rises again**.

■ REAL EXAMPLE — The radar “recovery” = harvest sign

Think of radar as someone feeling a field with their eyes closed. As long as wheat is dense and standing, their hand meets a “forest” of stems (low ratio). Once the field is harvested, their hand finds bare, smooth soil (rising ratio). This **ratio recovery** is therefore a direct signal that harvest occurred. In our data we call it the “recovery”.

■ KEY TAKEAWAY

Optical = COLOUR (green → yellow = senescence). Radar = STRUCTURE (dense → bare soil = harvest). Both point to the same event, but with different strengths and weaknesses. Combining them gives a more robust estimate — confirmed by scientific literature (wheat studies in Germany, Lebanon, France).

4. The derived-target recipe, step by step

Here is concretely how we compute a harvest date for each parcel. Think of it as a cooking recipe with several steps.

4.1 Blending the two signals intelligently

For a given parcel, we have two harvest-date estimates: one from optical data (NDVI senescence) and one from radar (ratio recovery). The question is: **how to combine them?**

Rather than a simple “50/50”, we use a **quality-weighted average**. The idea: trust the more reliable signal for that specific parcel more.

The radar signal's weight is its **quality indicator** (computed earlier), a value between 0 and 1. If the radar signal is clear and consistent, its weight is high (close to 1). If it is noisy or doubtful, its weight is low.

■ REAL EXAMPLE — Actual computation for one parcel (parcel 2020_10209024)

- Optical senescence detected: day 164 of the year (mid-June)
 - Radar recovery detected: day 194 (mid-July)
 - Optical weight = 1.0 (signal present); radar weight = 1.0 (high quality)
 - Raw estimate = $(1.0 \times 164 + 1.0 \times 194) / (1.0 + 1.0) = \text{day 179}$
- This parcel thus gets a raw estimate at day 179, before calibration.

4.2 Annual recalibration (the main safeguard)

Raw estimates may have a systematic offset (for example, senescence always comes a little before the actual harvest). To correct this, we apply an **annual calibration**: for each year, we shift all values by the same amount so that **the median of the parcels matches exactly the official Céré'Obs date** of that year.

■ REAL EXAMPLE — Calibration of year 2020

- Raw median of the 300 parcels of 2020: day 179
- Official Céré'Obs date 2020: day 185
- Shift to apply: $185 - 179 = +6 \text{ days}$

We thus add 6 days to all 2020 parcels. The parcel from the previous example (day 179) becomes day 185. The median then matches the official reference perfectly, while keeping the differences between early and late parcels.

This mechanism is clever: it combines the **best of both worlds**. Parcel-to-parcel variability comes from satellites (innovation), but the overall “level” stays anchored to the official regional truth (reliability).

4.3 The five safeguards, explained

An “invented” target built from signals carries risks. We therefore set up five protections (safeguards):

- **Annual calibration** — described above: ensures the mean matches the official reality.
- **Biological clipping** — we force all dates to stay within a realistic range (mid-June to mid-August). A harvest date in April would be absurd and is corrected.

- **Confidence weighting** — if both signals are missing or poor, we do not invent random values: we fall back to the regional date (fallback solution).
- **Exclusion of “cheating” variables** — crucial point detailed in section 6: we forbid the model from using the signals that built the target.
- **Consistency checks** — we verify the target is neither “flat” (too little variability = useless) nor “chaotic” (too much variability = noise).

5. Results: what we expected vs what we obtained

Before running the computations, we defined what would count as a “good” result. Here is the comparison between our expectations and the reality obtained.

Criterion	Expected (goal)	Obtained	Verdict
Parcel-to-parcel variability	2 to 25 days	9.0 days	Ideal
Number of distinct dates	more than 100	142	Met
Consistency with official (annual correlation)	above 0.9	1.000	Perfect
Median calibration	gap $\leq$ 2 days	0 days (5/5)	Perfect
Use of fallback solution	below 20%	1.4%	Excellent
Parcels using both sensors	majority	70.3%	Excellent

Table 2 — Validation criteria: goals vs results.

How to read these results?

The 9-day variability is the most important result. It means parcels have harvest dates spread typically over ± 9 days around the annual mean. This is **agronomically realistic**: within one region, some parcels (well-drained soil, southern exposure) ripen earlier than others. Zero variability would have meant our target adds nothing beyond the regional one; 40-day variability would have signalled incoherent noise.

The 1.000 correlation with the official source confirms our target perfectly respects the “ranking” of years: 2021 stays the latest year, 2022 stays early, etc. The year-to-year climate signal is fully preserved.

70% of parcels use both sensors simultaneously, validating our fusion strategy. Only 1.4% of parcels had to use the fallback (regional date), a negligible proportion.

Signal source distribution

Source used	# parcels	Share	Meaning
Both (optical+radar)	1055	70.3%	Ideal case
Optical only	400	26.7%	Radar absent/doubtful
Radar only	24	1.6%	Optical missing (clouds)
Fallback (regional)	21	1.4%	No usable signal

Table 3 — Where each parcel's estimate comes from.

6. The “circular reasoning” trap and how we avoid it

This is the most delicate methodological point of the whole project, and we explain it clearly because an examiner will pay attention to it.

The problem

We built the derived target from two signals: optical senescence and radar recovery. If we then gave these same signals to the model to predict the target, the model would just have to “redo the calculation” we performed ourselves. It would get an excellent score... but **completely misleading**, because it would have learned nothing useful. This is called **circular reasoning** (or data leakage).

■ REAL EXAMPLE — An analogy to understand circularity

It is like asking a student to guess the answer to an addition, but having already slipped the answer onto their sheet. They would get 20/20, but that score would prove nothing about their ability to calculate. To assess honestly, you must **remove the answer from their sheet** — exactly what we do by excluding the cheating variables.

Our solution

For the forecasting datasets, we **strictly forbid** 14 variables: all those that built the target, plus all those describing June-July events (hence after the forecast moment). The program automatically checks that none of these forbidden variables sneaks into the data — and the check passed successfully.

Examples of forbidden variables:

- **NDVI senescence** (date, slope, duration) — used to build the target.
- **Radar recovery** (date, quality) — used to build the target.
- **June statistics** — occur after the forecast moment.
- **April-June integrals** — include information posterior to the cutoff.

■ KEY TAKEAWAY

By excluding these variables, we accept the model will have a more modest score — but this score will be **HONEST** and **TRULY** predictive. It is a choice of scientific rigour: better a modest but true result than a brilliant but illusory one.

7. Forecast early or late? The two cutoff dates

A farmer wants to be warned as early as possible. But the earlier we forecast, the less information we have (the season is not over). There is therefore a **trade-off** between anticipation and accuracy. To study it, we created two dataset versions:

Version	Cutoff date	Anticipation	# variables
may31	May 31	≈ 6 weeks before	91 variables
june15	June 15	≈ 4 weeks before	107 variables

Table 4 — The two forecast horizons tested.

The principle: at the cutoff date, we keep only information **already available** at that moment. On May 31, June weather is not yet known; on June 15, it is. This is why the “june15” version has 16 more variables (mostly complete spring weather and the NDVI peak).

■ REAL EXAMPLE — Why compare the two?

If the “june15” model is clearly better than “may31”, it means the last two weeks of information (mid-May to mid-June) are valuable for forecasting. If both are equivalent, then we can forecast as early as May 31 with no loss — excellent news for farmers. This comparison is a scientific contribution in itself.

■ KEY TAKEAWAY

This “forecast-horizon” approach reproduces what real operational systems do (e.g. a yield forecasting system in France uses a mid-June cutoff). Our approach is thus aligned with the state of the art.

8. The five deliverables produced

At the end of this work, we have five complementary datasets:

#	Name	Target	Variables	Main use
1	master_raw_regional	Regional	Raw	Exploration
2	master_ml_regional	Regional	All	Reference (baseline)
3	master_raw_derived	Derived	Raw	Exploration
4a	master_ml_derived_may31	Derived	Before May 31	Operational forecast
4b	master_ml_derived_june15	Derived	Before June 15	Operational forecast

Table 5 — The five deliverables and their role.

This collection forms a rich **experimental design**. By comparing model performance across these datasets, the thesis can answer several questions: the effect of target granularity (regional vs parcel-level), the effect of forecast horizon (May 31 vs June 15), and the impact of the operational constraint (limited variables) on prediction quality.

9. Limitations and scientific honesty

■■ SCIENTIFIC HONESTY NOTE

The derived target is a PROXY (a constructed estimate), NOT field-measured ground truth. Unlike studies with field records (harvest date noted by an agronomist), our target is inferred from satellite signals themselves, then calibrated on the regional statistic. This proxy nature must be clearly acknowledged in the thesis and at the defence.

Identified and documented limitations:

- **No parcel-level ground truth** — to our knowledge, no field harvest-date records exist for our 1,500 parcels. The derived target therefore cannot be validated against an independent parcel-level reference.
- **3.1% of clipped parcels** — 47 parcels had an estimate outside the realistic range (very early senescence, possibly due to stress or poor signal) and were brought back to the lower bound.
- **Radar signal degraded some years** — in 2024 (wet spring) and 2021 (cool spring), radar signal quality was lower. The quality indicator accounts for this, but it remains a data limitation.
- **Department-level weather** — weather comes from 6 points (one per department), not each parcel. It mostly captures between-year variation, little fine variation between neighbouring parcels.

Acknowledging these limitations does not weaken the work: on the contrary, it demonstrates **methodological mastery**. The “regional” dataset remains a fully reliable basis, and the “derived” dataset is an honest, well-framed exploration of a more ambitious approach.

10. Glossary of technical terms

Term	Definition
NDVI	Vegetation index computed from optical imagery. Measures “how green and active vegetation is”. High = healthy plant; low = bare soil or dry plant.
EVI	Improved NDVI variant, less sensitive to soil and atmosphere.
NDWI	Index measuring vegetation water content. Useful to detect drying.
Senescence	Phase where the plant yellows and dries at the end of its cycle. Reflected by an NDVI drop. Precedes harvest by a few days to two weeks.
VV / VH	The two “polarisations” measured by Sentinel-1 radar. Their ratio (VV/VH) informs on vegetation structure.
VV/VH ratio	Radar structure indicator. Low when canopy is dense; rises when bare soil reappears (after harvest).
Recovery	Moment when the radar ratio rises again, signalling canopy degradation — often associated with harvest.
DOY	“Day Of Year”: day number within the year (1 = Jan 1, 185 ≈ July 3). A practical way to handle dates in computations.
Target	The value the model must learn to predict — here, the harvest date.
Proxy	A constructed estimate serving as a substitute for an unavailable direct measurement.
Calibration	Adjusting values to match a known reference (here, the official regional date).
Data leakage	Error where the model has access, directly or indirectly, to the answer it should guess. Yields misleading scores.
Circular reasoning	A special case of leakage: using to predict the target the same information that built that target.
GDD	“Growing Degree Days”: cumulative heat received by the plant. A good indicator of its progress through the growth cycle.
Céré'Obs	Official French source (FranceAgriMer) tracking cereal harvest progress, region by region, week by week.
LOYO	“Leave-One-Year-Out”: validation method where the model is trained on four years and tested on the fifth, in turn.

End of methodological report — Phase 4bis & 4ter. Document written for educational purposes to support the Master's thesis.